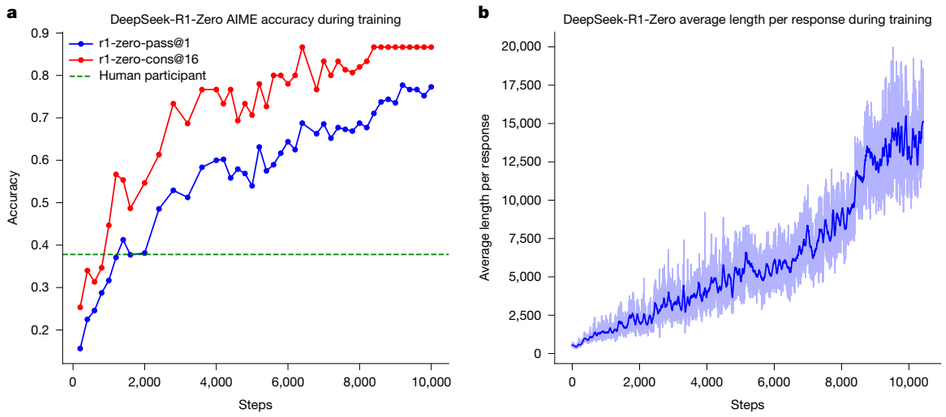


Fig. 1 | Accuracy and output length of DeepSeek-R1-Zero throughout the training process. a, AIME accuracy of DeepSeek-R1-Zero during training. AIME takes a mathematical problem as input and a number as output, illustrated in Extended Data Table 1. pass@1 and cons@16 are described in Supplementary Information, section 4.1. The baseline is the average score achieved by human participants in the AIME competition. **b**, The average response length of DeepSeek-R1-Zero on the training set during the RL process. DeepSeek-R1-Zero naturally learns to solve reasoning tasks with more thinking time. Note that a training step refers to a single policy update operation.

DeepSeek-R1-Zero

To implement large-scale RL of DeepSeek-R1-Zero, we use a highly efficient RL pipeline. Specifically, we use GRPO[9] as our RL algorithm, described in Methods section 'GRPO'. Furthermore, we use a rule-based reward system to compute accuracy and format rewards, with detailed methodologies outlined in Methods section 'Reward design'. Furthermore, our high-performance RL infrastructure is described in Supplementary Information, section 2.1, ensuring scalable and efficient training.

Specifically, we apply the RL technique on the DeepSeek-V3 Base to train DeepSeek-R1-Zero. During training, we design a straightforward template to require DeepSeek-R1-Zero to first produce a reasoning process, followed by the final answer. The prompt template is written as below.

"A conversation between User and Assistant. The User asks a question and the Assistant solves it. The Assistant first thinks about the reasoning process in the mind and then provides the User with the answer. The reasoning process and answer are enclosed within <think>...</think> and <answer>...</answer> tags, respectively, that is, <think> reasoning process here </think><answer> answer here </answer>. User: prompt. Assistant:", in which the prompt is replaced with the specific reasoning question during training. We intentionally limit our constraints to this structural format, avoiding any content-specific biases to ensure that we can accurately observe the natural progression of the model during the RL process.

Figure 1a shows the performance trajectory of DeepSeek-R1-Zero on the American Invitational Mathematics Examination (AIME) 2024 benchmark throughout the RL training process, in which the average pass@1 score on AIME 2024 shows a marked increase, jumping from an initial value of 15.6% to 77.9%. Also, by using the self-consistency decoding[10], the performance of the model can be further improved, achieving an accuracy of 86.7%. This performance greatly surpasses the average performance across all human competitors of the AIME. Besides the maths competitions, as shown in Supplementary Fig. 8, DeepSeek-R1-Zero also achieves remarkable performance in coding competitions and graduate-level biology, physics and chemistry problems. These results underscore the effectiveness of RL in enhancing the reasoning capabilities of LLMs.

As well as the progressive enhancement of reasoning capabilities during training, DeepSeek-R1-Zero also demonstrates self-evolutionary behaviour with RL training. As shown in Fig. 1b, DeepSeek-R1-Zero exhibits a steady increase in thinking time throughout training, driven only by intrinsic adaptation rather than external modifications. Making use of long CoT, the model progressively refines its reasoning, generating hundreds to thousands of tokens to explore and improve its problem-solving strategies.

The increase in thinking time helps with the autonomous development of sophisticated behaviours. Specifically, DeepSeek-R1-Zero increasingly exhibits advanced reasoning strategies such as reflective reasoning and systematic exploration of alternative solutions provided in Extended Data Fig. 1a, substantially boosting its performance on verifiable tasks such as maths and coding. Notably, during training, DeepSeek-R1-Zero exhibits an 'aha moment', shown in Table 1, characterized by a sudden increase in the use of the word 'wait' during reflections, provided in Extended Data Fig. 1b. This moment marks a distinct change in reasoning patterns and clearly shows the self-evolution process of DeepSeek-R1-Zero.

The self-evolution of DeepSeek-R1-Zero underscores the power and beauty of RL: rather than explicitly teaching the model how to solve a problem, we simply provide it with the right incentives and it autonomously develops advanced problem-solving strategies. This serves as a reminder of the potential of RL to unlock higher levels of capabilities in LLMs, paving the way for more autonomous and adaptive models in the future.

DeepSeek-R1

Although DeepSeek-R1-Zero exhibits strong reasoning capabilities, it faces several issues. DeepSeek-R1-Zero struggles with challenges such as poor readability and language mixing, as DeepSeek-V3 Base is trained on several languages, especially English and Chinese. To address these issues, we develop DeepSeek-R1, whose pipeline is illustrated in Fig. 2. In the initial stage, we collect thousands of cold-start data that exhibit a conversational, human-aligned thinking process, as detailed in Supplementary Information, section 2.3.2. RL training is then applied with hyperparameters in Methods section 'Training details of the first RL stage', data details in Supplementary Information, section 2.3.1, to improve the model performance with the

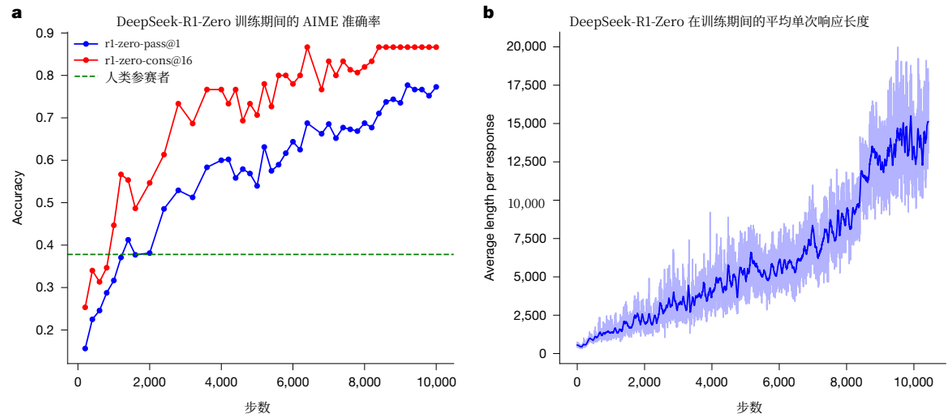


图 1 | DeepSeek-R1-Zero 在整个训练过程中的准确率和输出长度。 a，DeepSeek-R1-Zero 在训练期间的 AIME 准确率。AIME 以数学问题为输入，以一个数字为输出，如扩展数据表 1 所示。pass@1 和 cons@16 在补充信息第 4.1 节中进行了说明。基线为人类参赛者取得的平均分数这些参赛者参加了 AIME 竞赛。**b**，DeepSeek-R1-Zero 在 RL 过程中于训练集上的平均响应长度。DeepSeek-R1-Zero 自然地学会了通过更长的思考时间来解决推理任务。请注意，一个训练步指一次策略更新操作。

DeepSeek-R1-Zero

为实现 DeepSeek-R1-Zero 的大规模 RL，我们采用了高效的 RL 流水线。具体而言，我们使用 GRPO[9] 作为 RL 算法，该算法在方法部分的"GRPO"一节中进行了介绍。此外，我们使用基于规则的奖励系统来计算准确性奖励和格式奖励，详细方法见方法部分的"奖励设计"一节。此外，我们的高性能 RL 基础设施在补充信息第 2.1 节中进行了介绍，可确保训练的可扩展性和效率。

具体而言，我们将 RL 技术应用于 DeepSeek-V3 Base，以训练 DeepSeek-R1-Zero。在训练过程中，我们设计了一个简单的模板，要求 DeepSeek-R1-Zero 先生成推理过程，再给出最终答案。提示模板如下。

"用户与助手之间的对话。用户提出问题，助手解决问题。助手先在心中思考推理过程，然后向用户提供答案。推理过程和答案分别包含在 <think>...</think>and <answer>...</answer> tags 中，即 <think> reasoning 此处为过程 </think><answer> answer 此处 </answer>. 用户：提示。助手：" ，其中，训练时会将提示替换为具体的推理问题。我们有意将约束限定在这一结构格式上，避免任何针对特定内容的偏向，以确保能够准确观察模型在 RL 过程中的自然演进。

图 **1a** 展示了 DeepSeek-R1-Zero 在整个 RL 训练过程中在美国数学邀请赛（AIME）2024 基准上的性能变化轨迹，其中 AIME 2024 的平均 pass@1 得分显著提高，从初始值 15.6% 跃升至 77.9%。此外，通过使用自一致性解码[10]，，模型的性能可以进一步提升，达到 86.7% 的准确率。这一表现远超 AIME 全体人类参赛者的平均表现。除数学竞赛外，如补充图 8 所示，DeepSeek-R1-Zero 在编程竞赛以及研究生水平的生物学、物理学和化学问题上也取得了出色的表现。这些结果凸显了 RL 在增强 LLMs 推理能力方面的有效性。

除了在训练过程中推理能力逐步增强之外，DeepSeek-R1-Zero 在 RL 训练中还表现出自我演化行为。如图 **1b** 所示，DeepSeek-R1-Zero 的思考时间在整个训练过程中稳步增加，这一变化仅由内在适应驱动，而非外部修改。通过利用长 CoT，模型逐步完善其推理，生成数百至数千个词元来探索和改进其问题求解策略。

思考时间的增加有助于自主形成复杂行为。具体而言，DeepSeek-R1-Zero 越来越多地展现出高级推理策略，例如扩展数据图 1a 中所示的反思性推理和对替代解决方案的系统性探索，从而大幅提升其在数学和编程等可验证任务上的表现。值得注意的是，在训练过程中，DeepSeek-R1-Zero 出现了表 **1** 所示的'顿悟时刻'，其特征是在反思过程中使用'等等'一词的频率突然增加，如扩展数据图 1b 所示。这一时刻标志着推理模式的显著变化，并清晰地展现了 DeepSeek-R1-Zero 的自我演化过程。

DeepSeek-R1-Zero 的自我演化凸显了 RL 的力量与美妙：我们只需为模型提供恰当的激励，它就能自主形成高级问题求解策略，而无需明确教它如何解决问题。这提醒我们，RL 具有释放 LLMs 更高水平能力的潜力，为未来更具自主性和适应性的模型铺平了道路。

DeepSeek-R1

尽管 DeepSeek-R1-Zero 展现出强大的推理能力，但它仍面临一些问题。由于 DeepSeek-V3 Base 使用多种语言进行训练，尤其是英语和中文，DeepSeek-R1-Zero 存在可读性差和语言混杂等问题。为解决这些问题，我们开发了 DeepSeek-R1，其流程如图 **2** 所示。在初始阶段，我们收集了数千条冷启动数据，这些数据展现出对话式、与人类思维相契合的思考过程，详见补充信息第 2.3.2 节。随后进行 RL 训练，超参数见方法部分的"第一阶段 RL 训练细节"，数据详情见补充信息第 2.3.1 节，以通过以下方式提升模型性能：

Methods

GRPO

GRPO[9] is the RL algorithm that we use to train DeepSeek-R1-Zero and DeepSeek-R1. It was originally proposed to simplify the training process and reduce the resource consumption of proximal policy optimization (PPO)[31], which is widely used in the RL stage of LLMs[32]. The pipeline of GRPO is shown in Extended Data Fig. 2.

For each question q, GRPO samples a group of outputs $\{o_1, o_2, \ldots, o_G\}$ from the old policy $\pi_{\theta_{\text{old}}}$ and then optimizes the policy model π_θ by maximizing the following objective:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{\text{old}}}(O|q)]$$
$$\frac{1}{G}\sum_{i=1}^{G}\left(\min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)}A_i,\ \text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)}, 1-\epsilon, 1+\epsilon\right)A_i\right)\right. \quad (1)$$
$$\left. - \beta\mathbb{D}_{KL}(\pi_\theta||\pi_{\text{ref}})\right),$$

$$\mathbb{D}_{\text{KL}}(\pi_\theta||\pi_{\text{ref}}) = \frac{\pi_{\text{ref}}(o_i|q)}{\pi_\theta(o_i|q)} - \log\frac{\pi_{\text{ref}}(o_i|q)}{\pi_\theta(o_i|q)} - 1, \quad (2)$$

in which π_{ref} is a reference policy, ϵ and β are hyperparameters and A_i is the advantage, computed using a group of rewards $\{r_1, r_2, \ldots, r_G\}$ corresponding to the outputs in each group:

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \cdots, r_G\})}{\text{std}(\{r_1, r_2, \cdots, r_G\})}. \quad (3)$$

We give a comparison of GRPO and PPO in Supplementary Information, section 1.3.

Reward design

The reward is the source of the training signal, which decides the direction of RL optimization. For DeepSeek-R1-Zero, we use rule-based rewards to deliver precise feedback for data in mathematical, coding and logical reasoning domains. For DeepSeek-R1, we extend this approach by incorporating both rule-based rewards for reasoning-oriented data and model-based rewards for general data, thereby enhancing the adaptability of the learning process across diverse domains.

Rule-based rewards. Our rule-based reward system mainly consists of two types of reward: accuracy rewards and format rewards.

Accuracy rewards evaluate whether the response is correct. For example, in the case of maths problems with deterministic results, the model is required to provide the final answer in a specified format (for example, within a box), enabling reliable rule-based verification of correctness. Similarly, for code competition prompts, a compiler can be used to evaluate the responses of the model against a suite of predefined test cases, thereby generating objective feedback on correctness.

Format rewards complement the accuracy reward model by enforcing specific formatting requirements. In particular, the model is incentivized to encapsulate its reasoning process within designated tags, specifically <think> and </think>. This ensures that the thought process of the model is explicitly delineated, enhancing interpretability and facilitating subsequent analysis.

$$\text{Reward}_{\text{rule}} = \text{Reward}_{\text{acc}} + \text{Reward}_{\text{format}} \quad (4)$$

The accuracy, reward and format reward are combined with the same weight. Notably, we abstain from applying neural reward models—whether outcome-based or process-based-to reasoning tasks. This decision is predicated on our observation that neural reward models are susceptible to reward hacking during large-scale RL. Moreover, retraining such models necessitates substantial computational resources and introduces further complexity into the training pipeline, thereby complicating the overall optimization process.

Model-based rewards. For general data, we resort to reward models to capture human preferences in complex and nuanced scenarios. We build on the DeepSeek-V3 pipeline and use a similar distribution of preference pairs and training prompts. For helpfulness, we focus exclusively on the final summary, ensuring that the assessment emphasizes the use and relevance of the response to the user while minimizing interference with the underlying reasoning process. For harmlessness, we evaluate the entire response of the model, including both the reasoning process and the summary, to identify and mitigate any potential risks, biases or harmful content that may arise during the generation process.

Helpful reward model. For helpful reward model training, we first generate preference pairs by prompting DeepSeek-V3 using the Arena-Hard prompt format, listed in Supplementary Information, section 2.2, for which each pair consists of a user query along with two candidate responses. For each preference pair, we query DeepSeek-V3 four times, randomly assigning the responses as either Response A or Response B to mitigate positional bias. The final preference score is determined by averaging the four independent judgments, retaining only those pairs for which the score difference (Δ) exceeds 1 to ensure meaningful distinctions. Furthermore, to minimize length-related biases, we ensure that the chosen and rejected responses of the whole dataset have comparable lengths. In total, we curated 66,000 data pairs for training the reward model. The prompts used in this dataset are all non-reasoning questions and are sourced either from publicly available open-source datasets or from users who have explicitly consented to share their data for the purpose of model improvement. The architecture of our reward model is consistent with that of DeepSeek-R1, with the addition of a reward head designed to predict scalar preference scores.

$$\text{Reward}_{\text{helpful}} = \text{RM}_{\text{helpful}}(\text{Response}_A, \text{Response}_B) \quad (5)$$

The helpful reward models were trained with a batch size of 256, a learning rate of 6×10^{-6} and for a single epoch over the training dataset. The maximum sequence length during training is set to 8,192 tokens, whereas no explicit limit is imposed during reward model inference.

Safety reward model. To assess and improve model safety, we curated a dataset of 106,000 prompts with model-generated responses annotated as 'safe' or 'unsafe' according to predefined safety guidelines. Unlike the pairwise loss used in the helpfulness reward model, the safety reward model was trained using a pointwise methodology to distinguish between safe and unsafe responses. The training hyperparameters are the same as the helpful reward model.

$$\text{Reward}_{\text{safety}} = \text{RM}_{\text{safety}}(\text{Response}) \quad (6)$$

For general queries, each instance is categorized as belonging to either the safety dataset or the helpfulness dataset. The general reward, $\text{Reward}_{\text{general}}$, assigned to each query corresponds to the respective reward defined in the associated dataset.

Training details

Training details of DeepSeek-R1-Zero. To train DeepSeek-R1-Zero, we set the learning rate to 3×10^{-6}, the Kullback–Leibler (KL) coefficient to 0.001 and the sampling temperature to 1 for rollout. For each question, we sample 16 outputs with a maximum length of 32,768 tokens before the 8.2k step and 65,536 tokens afterward. As a result, both the performance and response length of DeepSeek-R1-Zero exhibit a substantial jump at the 8.2k step, with training continuing for a total of 10,400 steps, corresponding to 1.6 training epochs. Each training step consists of 32 unique questions, resulting in a training batch size of 512 per step. Every 400 steps, we replace the reference model with

方法

GRPO

GRPO[9] 是我们用于训练 DeepSeek-R1-Zero 和 DeepSeek-R1 的 RL 算法。该算法最初被提出，旨在简化近端策略优化（PPO）的训练过程并降低其资源消耗[31]，，PPO 广泛用于 LLM 的 RL 阶段[32]。GRPO 的流程如扩展数据图 2 所示。

对于每个问题 q，GRPO 从旧策略中采样一组输出 $\{o_1, o_2, \ldots, o_G\}$，旧策略为 $\pi_{\theta_{\text{old}}}$，然后通过最大化以下目标来优化策略模型 π_θ：

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{\text{old}}}(O|q)]$$
$$\frac{1}{G}\sum_{i=1}^{G}\left(\min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)}A_i,\ \text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)}, 1-\epsilon, 1+\epsilon\right)A_i\right)\right. \quad (1)$$
$$\left. - \beta\mathbb{D}_{KL}(\pi_\theta||\pi_{\text{ref}})\right),$$

$$\mathbb{D}_{\text{KL}}(\pi_\theta||\pi_{\text{ref}}) = \frac{\pi_{\text{ref}}(o_i|q)}{\pi_\theta(o_i|q)} - \log\frac{\pi_{\text{ref}}(o_i|q)}{\pi_\theta(o_i|q)} - 1, \quad (2)$$

其中，π_{ref} 为参考策略，ϵ 和 β 为超参数，A_i 为优势值，使用与每组输出对应的一组奖励 $\{r_1, r_2, \ldots, r_G\}$ 计算得到：

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \cdots, r_G\})}{\text{std}(\{r_1, r_2, \cdots, r_G\})}. \quad (3)$$

我们在补充信息第 1.3 节中给出了 GRPO 与 PPO 的比较。

奖励设计

奖励是训练信号的来源，决定了 RL 优化的方向。对于 DeepSeek-R1-Zero，我们使用基于规则的奖励，为数学、编程和逻辑推理领域的数据提供精确反馈。对于 DeepSeek-R1，我们扩展了这一方法，将面向推理的数据所用的基于规则的奖励与通用数据所用的基于模型的奖励相结合，从而增强学习过程在不同领域中的适应性。

基于规则的奖励。我们基于规则的奖励系统主要包含两类奖励：准确性奖励和格式奖励。

准确性奖励评估回答是否正确。例如，对于结果确定的数学问题，要求模型以指定格式（例如，置于方框内）给出最终答案，以便基于规则可靠地验证其正确性。同样，对于编程竞赛提示，可以使用编译器，根据一组预定义的测试用例评估模型的回答，从而生成关于正确性的客观反馈。

格式奖励通过强制执行特定的格式要求，对准确性奖励模型起到补充作用。具体而言，模型受到激励，将其推理过程封装在指定标签内，即 <think> and </think>. 这确保了模型的思考过程得到明确标示，从而增强可解释性，并便于后续分析。

$$\text{Reward}_{\text{rule}} = \text{Reward}_{\text{acc}} + \text{Reward}_{\text{format}} \quad (4)$$

准确性奖励和格式奖励以相同的权重组合。值得注意的是，我们不对推理任务应用神经奖励模型，无论是基于结果的还是基于过程的。这一决定基于我们的观察：神经奖励模型在大规模 RL 过程中容易受到奖励破解的影响。

此外，重新训练此类模型需要大量计算资源，并会增加训练流程的复杂性，从而使整个优化过程更加复杂。

基于模型的奖励。对于通用数据，我们采用奖励模型来捕捉复杂且细微场景中的人类偏好。我们基于 DeepSeek-V3 流程，使用分布相似的偏好对和训练提示。对于有用性，我们仅关注最终总结，确保评估着重考察回答对用户的实用性和相关性，同时尽量减少对底层推理过程的干扰。对于无害性，我们评估模型的整个回答，包括推理过程和总结，以识别并减轻生成过程中可能出现的任何潜在风险、偏见或有害内容。

有用性奖励模型。在训练有用性奖励模型时，我们首先使用补充信息第 2.2 节中列出的 Arena-Hard 提示格式向 DeepSeek-V3 提供提示，生成偏好对，每对包含一个用户查询和两个候选回答。对于每个偏好对，我们向 DeepSeek-V3 查询四次，将回答随机分配为 Response A 或 Response B，以减轻位置偏差。最终偏好分数通过对四次独立判断取平均值来确定，仅保留分数差（Δ）超过 1 的偏好对，以确保差异具有意义。此外，为尽量减少与长度相关的偏差，我们确保整个数据集中被选中的回答和被拒绝的回答长度相近。我们总共整理了 66,000 个数据对用于训练奖励模型。该数据集中使用的提示均为非推理问题，来源于公开可用的开源数据集，或明确同意共享其数据以改进模型的用户。我们的奖励模型架构与 DeepSeek-R1 一致，并额外增加了一个用于预测标量偏好分数的奖励头。

$$\text{Reward}_{\text{helpful}} = \text{RM}_{\text{helpful}}(\text{Response}_A, \text{Response}_B) \quad (5)$$

有用性奖励模型以 256 的批量大小、6×10^{-6} 的学习率在训练数据集上训练一个轮次。训练期间的最大序列长度设为 8,192 个 token，而奖励模型推理期间不设明确的长度限制。

安全性奖励模型。为了评估和提高模型的安全性，我们整理了一个包含 106,000 条提示的数据集，并根据预先定义的安全准则，将模型生成的回答标注为“安全”或“不安全”。与有用性奖励模型所使用的成对损失不同，安全性奖励模型采用逐点方法进行训练，以区分安全与不安全的回答。训练超参数与有用性奖励模型相同。

$$\text{Reward}_{\text{safety}} = \text{RM}_{\text{safety}}(\text{Response}) \quad (6)$$

对于一般查询，每个实例被归类为属于安全性数据集或有用性数据集。分配给每个查询的通用奖励 $\text{Reward}_{\text{general}}$ 对应于其所属数据集中定义的相应奖励。

训练细节

DeepSeek-R1-Zero 的训练细节。为训练 DeepSeek-R1-Zero，我们将学习率设为 3×10^{-6}，将 Kullback–Leibler（KL）系数设为 0.001，并将 rollout 的采样温度设为 1。对于每个问题，我们采样 16 个输出，在第 8.2k 步之前，最大长度为 32,768 个 token，此后为 65,536 个 token。因此，DeepSeek-R1-Zero 的性能和回答长度均在第 8.2k 步出现显著跃升，训练持续进行至总计 10,400 步，对应 1.6 个训练轮次。每个训练步骤包含 32 个不同的问题，因此每步的训练批量大小为 512。每隔 400 步，我们将参考模型替换为